Probability & Statistics

Basic terms in Probability ÷

1. Random Experiment :
An action whose outcome is not known beforehand.
e.g. Rolling a die.

2. Outcome :
A single possible result of an expt.
e.g. Getting 4 when rolling a die.

3. Sample search (S) :
The set of all possible outcomes.
$S = \{1, 2, 3, 4, 5, 6\}$

4. Event (E) :
A set of one or more outcomes that satisfy a condition.
e.g. Getting an even number.
$E = \{2, 4, 6\}$

5. favourable outcome :
Outcome that satisfy the event we are interested in –
e.g. for the event "getting an even number"
$n(E) = 3$
↓
nos of favourable outcome.

6. Equally likely outcomes :
outcomes that have same probability of occuring.
e.g. for a die $P(1) = P(2) = \dots P(6) = \frac{1}{6}$

7. probability $= \frac{n(E)}{n(S)} = \frac{\text{favourable outcomes}}{\text{Total number of outcomes}}$

1. Independent Events ÷

$$\boxed{P(E \cap F) = P(E) \times P(F)}$$

E and F happen together, and the occurence of one doesn't affect the probability of other

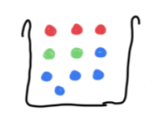
e.g. tossing two coin simultaneously
- Event E: first coin shows head
- Event F: second coin shows head

since, the coins are independent:
$P(E) = \frac{1}{2}$; $P(F) = \frac{1}{2}$

prob. of getting HEAD on Both coins :
$P(E \cap F) = P(E) \times P(F) = \frac{1}{4}$

sample space : {HH, HT, TH, TT}
only HH satisfies both events

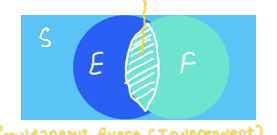
eg②.



We drew one Ball, notes it's color & put it Back
↓
With replacement, the total nos of Balls & color count remains unchanged
So, draws are independent.

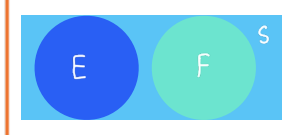
prob. (getting RED first AND GREEN second)
$\Rightarrow \left(\frac{3}{10}\right)\left(\frac{2}{10}\right) \Rightarrow \frac{3}{50}$

- something is common.
- $E \cap F$; E AND F



Simultaneous Event (Independent)

2.) Mutually Exclusive Events ÷



- $P(E \cap F) = 0$;
- $P(E \cup F) = P(E) + P(F) - \underset{=0}{P(E \cap F)}$
  $\Rightarrow P(E) + P(F)$

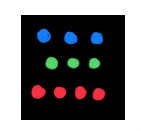
- Simultaneous, event can occur at same time (whether dependent or independent).
- Mutually Exclusive events can't happen together If one event happens the other can not.

e.g. ① two coin tosses
$S = \{HH, HT, TH, TT\}$
Consider :
E: first coin shows Head = {HH, HT}
F: first coin shows Tail = {TH, TT}
Can E & F happen togethe ? No

$P(E \cup F) = P(E) + P(F) - \underset{(\approx 0)}{P(E \cap F)} \rightarrow \frac{1}{2} + \frac{1}{2} \Rightarrow 1$

Notice these events are also exhaustive because one of them must happen.

3.) Dependent Events ÷ {Sequential Events}



Box

P (4 red Ball is drawn at random taken one at a time) ⇒ ?

→ $P(R_1) = \frac{n(R)}{n(S)} = \frac{4}{10}$ (Ist red ball is drawn)

→ $P\left(\frac{R_2}{R_1}\right) = \frac{3}{9}$ (IInd Red ball is drawn)

→ $P\left(\frac{R_3}{R_1, R_2}\right)$ or $P\left(\frac{R_3}{R_1 \cap R_2}\right)$ (IIIrd red Ball is drawn)
$\Rightarrow 2/8$

→ $P\left(\frac{R_4}{R_1, R_2, R_3}\right) \Rightarrow \frac{1}{7}$ (IVth Red ball is drawn)

Prob. of drawing 4th Red Ball, given 3 red Ball is already drawn :

$$P(R_1 \cap R_2 \cap R_3 \cap R_4) \Rightarrow P(R_1) \cdot P\left(\frac{R_2}{R_1}\right) \cdot P\left(\frac{R_3}{R_1 \cap R_2}\right) \cdot P\left(\frac{R_4}{R_1 \cap R_2 \cap R_3}\right)$$

2nd draw dependent on 1st draw
3rd draw dependent on 2nd draw

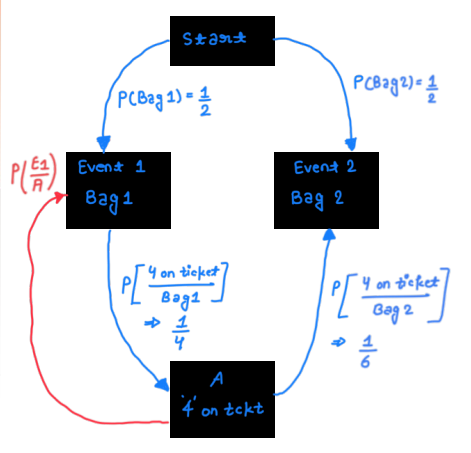
$$\boxed{P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right)} \rightarrow \text{W/O replacement / Dependent Events}$$

$$\boxed{\underbrace{P\left(\frac{B}{A}\right)}_{\text{Conditional Probability}} = \frac{P(A \cap B)}{P(A)}} \rightarrow \text{Both happen}$$

4.) Baye's Theorem ÷

Ques). A Bag contains 4 tickets numbered [1, 2, 3, 4] and another bag contains 6 tickets [2, 4, 6, 7, 8, 9].

One Bag is chosen and a ticket is drawn.
The probability that the ticket bears number 4 is —

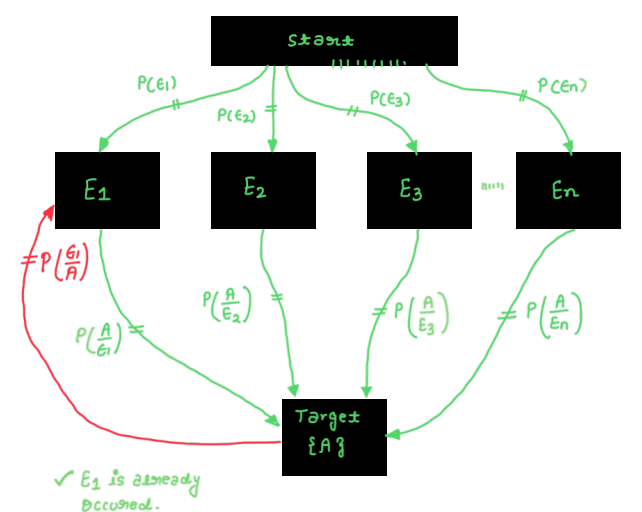


P (4 on ticket) = ?

$$P(\text{bag } 1) \cdot P\left(\frac{4 \text{ on tckt}}{\text{bag } 1}\right) + P(\text{bag } 2) \cdot P\left(\frac{4 \text{ on tckt}}{\text{bag } 2}\right)$$
⇓
$$P(A) = \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{6} \Rightarrow 5/24$$
→ prob. ticket bears number '4'.

$$P\left(\frac{E_1}{A}\right) = P\left(\frac{\text{bag } 1}{4 \text{ on tckt}}\right) \Rightarrow \frac{\frac{1}{2} \times \frac{1}{4}}{5/24} \Rightarrow \frac{3}{5}$$
↓
given '4' on ticket, find probability that it's from Bag 1.

$$P\left(\frac{E_2}{A}\right) = P\left(\frac{\text{bag } 2}{4 \text{ on tckt}}\right) \Rightarrow \frac{\frac{1}{2} \times \frac{1}{6}}{5/24} \Rightarrow \frac{1 \times 24}{12 \times 5} \Rightarrow \frac{2}{5}$$



✓ E_1 is already occured.
✓ $\frac{A}{E_1}$ is occuring

[$E_1, E_2, E_3, \dots, E_n$ are mutually Exclusive Events].

$$\boxed{P(A) = P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2) \cdot P\left(\frac{A}{E_2}\right) + \dots + P(E_n) \cdot P\left(\frac{A}{E_n}\right)}$$

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1) \cdot P\left(\frac{A}{E_1}\right)}{\sum_{i=1}^{n} P(E_i) \cdot P\left(\frac{A}{E_i}\right)}$$

Law of Total probability